

Linear Algebra (Inter)

Name: _____

2024 Fall

Final Exam

2025.1.4

Exam Duration: 3 Hours

Student ID: _____

This exam includes 11 pages (including this page) and 5 problems. Please check to see if there is any missing page, and then write down your name and student ID number on this page and the first page of your answer sheets. Also write down the initials of your name on the top of every page of your answer sheets, in case they are scattered.

This exam is open book. You are allowed to consult your textbook and notes, and elementary calculators are fine (those without internet connections). Plagiarism of all kinds are strictly forbidden and will be severely punished.

Please write down your answers to the problems in the **SEPARATE ANSWER SHEETS**, and follow the following rules:

- **Always explain your answer.** You should always explain your answers. Any problem answered with nothing but a single answer would receive no credit. If you copied the result from pre-calculations (since we have an open portion for this exam), just write “by pre-calculation”. If you spot some results without any calculation, just write “I can spot that blah blah”.
- **Write cleanly and legible.** Make sure that your writings can be read. The graders are NOT responsible to decipher illegible writings.
- **Partial credits will be given.**
- Blank spaces are provided in the exams. Feel free to use them as scratch papers. However, your formal answer has to be written in the **SEPARATE ANSWER SHEETS**, as required by the University.
- The total score of the exam is 50. If your total score exceeds 50 (there are 54 points in total), it will be recorded as 50.

Problem	Points	Score
1	10	
2	11	
3	13	
4	9	
5	11	
Total:	54	

1. Consider the matrix $A = \begin{bmatrix} 2 & i & -i \\ -i & 2 & i \\ i & -i & 2 \end{bmatrix}$. This matrix might describe some system in quantum mechanics, where the eigenvalues are energies, and the eigenvectors are stable states of the system.

The subproblems will be studying variations of A in a polynomial manner. Obviously eigen-informations and spectral theorems will be relevant.

(a) (2 points) Find $\det(2A - 2I)$.

(b) (3 points) Find $\det \begin{bmatrix} A & iA \\ -iA & -A \end{bmatrix}$.

(c) (3 points) Find all eigenvalues and all eigenvectors of $(A - 2I)^2$.

(d) (2 points) Find the largest and smallest possible value of $\mathbf{v}^* A \mathbf{v}$ among all UNIT vectors $\mathbf{v} \in \mathbb{C}^n$.

Answer:

Note that A has eigenvalue $2, 2 - \sqrt{3}, 2 + \sqrt{3}$. The eigenvectors are the span of

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \text{ of } \begin{bmatrix} \sqrt{3} + i \\ -\sqrt{3} + i \\ -2i \end{bmatrix}, \text{ and of } \begin{bmatrix} \sqrt{3} - i \\ -\sqrt{3} - i \\ 2i \end{bmatrix}.$$

1. -16 .

2. -32 . Block elimination gives $\begin{bmatrix} A & iA \\ -iA & -A \end{bmatrix} = \begin{bmatrix} I & \\ -iI & I \end{bmatrix} \begin{bmatrix} A & \\ & -2A \end{bmatrix} \begin{bmatrix} I & iI \\ & I \end{bmatrix}$, so the answer is $\det(A) \det(-2A)$.

3. $0, 3, 3$. span of $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, and span of $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$.

4. $2 - \sqrt{3}, 2 + \sqrt{3}$.

2. If the weather in one day has probability p to be sunny, probability q to be cloudy, and probability $1-p-q$ to be rainy, then we say the weather probability distribution of this day is $\begin{bmatrix} p \\ q \\ 1-p-q \end{bmatrix}$. Given the weather probability distribution in a day $\mathbf{x}$, the weather probability distribution in the next day will be $A\mathbf{x}$, where we have

$$A = \begin{bmatrix} 0.9 & 0.1 & 0 \\ 0.1 & 0.8 & 0.1 \\ 0 & 0.1 & 0.9 \end{bmatrix}.$$

The subproblems will try to understand this dynamical system and how it may evolve.

- (a) (4 points) Find all eigenvalues and eigenspaces of $A - I$. Is $A - I$ invertible? diagonalizable?
- (b) (3 points) Find the formula for A^n . (You may write the answer as a product of matrices and inverses without calculating the entries and you may leave exponential expressions as part of the answer.)
- (c) (2 points) Over a very long period of time, what is the percentage of rainy days?
- (d) (2 points) Find a polynomial $p(x)$ such that $p(A) = O$. (You may leave your answer as a product of factor polynomials.)

Answer:

1. $0, -0.1, -0.3$, not invertible, diagonalizable. Eigenspaces are the three spans of the columns of $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & -2 \\ 1 & -1 & 1 \end{bmatrix}$.
2. $X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & -2 \\ 1 & -1 & 1 \end{bmatrix}$, and $D = \begin{bmatrix} 1 & & \\ & 0.9^n & \\ & & 0.7^n \end{bmatrix}$, and $A^n = XDX^{-1}$.
3. $\frac{1}{3}$ of the days are rainy.
4. $p(x) = (x - 1)(x - 0.9)(x - 0.7)$.

3. We want to solve the equation $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 0 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix}.$$

The subproblems will try to solve this system, but something fishy is going on.

- (a) (3 points) Find the RREF of the 3×3 augmented matrix $[A \ \mathbf{b}]$, and find all solutions to $A\mathbf{x} = \mathbf{b}$. (Here A provides the first two columns, and $\mathbf{b}$ is the third column.)

- (b) (3 points) Are AA^T and $A^T A$ positive definite? Positive semidefinite? Negative definite? Negative semidefinite? Indefinite?

- (c) (4 points) Find the orthogonal projection to $\text{Ran}(2A)$ and $\text{Ker}(AA^T)$.

- (d) (3 points) Find the least square solution to $A\mathbf{x} = 2\mathbf{b}$.

Answer:

1. RREF is identity, and there is no solution.
2. A is linearly independent, so $A^T A$ is positive definite, and AA^T is positive semidefinite.
3. $\frac{1}{6} \begin{bmatrix} 2 & 2 & 2 \\ 2 & 5 & -1 \\ 2 & -1 & 5 \end{bmatrix}$ and $\frac{1}{6} \begin{bmatrix} 4 & -2 & -2 \\ -2 & 1 & 1 \\ -2 & 1 & 1 \end{bmatrix}$.
4. $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$.

4. Consider this matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 2 & \\ & & 3 \end{bmatrix} \begin{bmatrix} 2 & 2 & -1 \\ -1 & 2 & 2 \\ 2 & -1 & 2 \end{bmatrix}.$$

This looks like SVD but not really??? The subproblems will try to do some related analysis.

(a) (3 points) If the SVD for $2A$ is $U\Sigma V^T$, find U, Σ, V .

(b) (3 points) If S is a unit sphere, Then $A(S)$ is an ellipsoid. Describe the lengths and directions of the three half-axes of this ellipsoid.

(c) (3 points) Find the spectral decomposition of $A^T A$.

Answer:

1. $U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \end{bmatrix}, V = \frac{1}{3} \begin{bmatrix} 2 & -1 & 2 \\ 2 & 2 & -1 \\ -1 & 2 & 2 \end{bmatrix}, \text{ and } \Sigma = \begin{bmatrix} 6\sqrt{3} & & \\ & 12\sqrt{2} & \\ & & 18\sqrt{6} \end{bmatrix}.$

2. The directions are the three columns of U , and the corresponding half-axes lengths are the singular values.

3. $A^T A = V D V^T$ where $D = \begin{bmatrix} 27 & & \\ & 72 & \\ & & 486 \end{bmatrix}.$

5. Let V be an abstract space of functions. We have a basis for V , $\mathcal{B} = (e^x, xe^x, x^2e^x, e^{2x})$. We have a linear transformation $L : V \rightarrow V$, defined as $L(f) = f'' - 3f' + 2f$.

The subproblems will study this abstract map over abstract vector spaces. We will change basis for some subproblems.

- (a) (3 points) Find the matrix for L under the basis $\mathcal{B}$.

- (b) (3 points) Find the rank of L^2 , and find a basis for its Kernel. (When you write vectors in this basis, do NOT write coordinates. I want the actual functions in V that span this subspace.)

- (c) (3 points) We have another basis for V , $\mathcal{C} = (x^2e^x, (2 - 2x)e^x, 2e^x, e^{2x})$. Find L under the basis $\mathcal{C}$ for its domain and codomain.

- (d) (2 points) Is L diagonalizable? Why? Is L^2 diagonalizable? Why?

Answer:

$$1. L = \begin{bmatrix} 0 & -1 & 2 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

$$2. L^2 = \begin{bmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \text{ So rank is 1, kernel basis is } e^x, xe^x, e^{2x}.$$

$$3. L = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \text{ Change of basis is } I_{B \leftarrow C} = \begin{bmatrix} 0 & 2 & 2 & 0 \\ 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \text{ and inverse}$$

$$\text{is } \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

4. Both are not diagonalizable. For L , $m_a(0) = 4$ while $m_g(0) = 2$. For L^2 , $m_a(0) = 4$ while $m_g(0) = 3$.